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Module 1: fundamental of microprocessor

- Fundamental of microprocessor architecture
- 8-bit microprocessor and microcontroller architecture
- Comparison of 8-bit microcontrollers
- 16-bit and 32-bit microcontrollers
- Definition of embedded system and its characteristics
- Role of microcontroller sin embedded systems
- Overview of the 8051 family

Module 2: the 8051 architecture

- Internal block diagram
- CPU
- ALU
- Address
- Data and control bus
- Working registers
- SFRs
- Clock and RESET circuits
- Stack and stack pointer
- Program counter
- I/O ports
- Memory structures
- Data and program memory
- Timing diagram and execution cycles

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Module 3: instruction set and programming

- Addressing modes
- Introduction
- Instruction syntax
- Data types
- Subroutines immediate addressing
- Register addressing
- Direct addressing
- Indirect addressing
- Relative addressing
- Indexed addressing
- Bit inherent addressing
- Bit direct addressing
- 8051 instruction set
- Instruction timings
- Data transfer instructions
- Arithmetic instructions
- Logical instructions
- Branch instruction
- Subroutine instructions
- Bit manipulation instruction
- Assembly language program
- C language program
- Assemblers and compilers
- Programming and debugging tools

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Module 4: memory and I/O interfacing

- Memory and I/O expansion buses
- Control signals
- Memory wait states
- Interfacing of peripherical
- Devices such as general purpose I/O
- ADC
- DAC
- Timers
- Counters
- Memory devices

Module 5: external communication interface

- Synchronous and asynchronous communication
- RS232
- SPI
- I2C
- Introduction and interfacing to protocols like blue -tooth and zig-bee

Module 6: applications

- LED
- LCD and keyboard interfacing
- Stepper motor interfacing
- DC motor interfacing
- Sensor interfacing